

CSCE330401 – Spring 2026

Digital Design 2

Project 1

**Standard Cell Library Design
and
Characterization (SKY130)**

Dr. Mohamed Shaalan

Project Report

Joudy El Gayar 900222142

Department of Computer Science and Engineering, AUC

1. Introduction

This report documents the design and SPICE-based NLDM (Non-Linear Delay Model) characterization of a 13-cell standard cell library using the SkyWater SKY130 open-source 180 nm process technology. All simulations were performed at the Typical-Typical (TT) corner, 25°C, and a supply voltage of 1.8 V.

The library consists of the following cell types:

- Inverter family: invx1, invx2, invx4, invx8
- 2-input NAND family: nand2x1, nand2x2, nand2x4
- 2-input NOR family: nor2x1, nor2x2, nor2x4
- 3-input majority gate family: maj3x1, maj3x2, maj3x4

For each cell, four 7x7 NLDM tables were extracted using automated SPICE simulation with ngspice: cell_rise, cell_fall, rise_transition, and fall_transition. The tables are indexed by input transition time (7 values, 0.0100 ns to 1.5000 ns) and output load capacitance (7 values, 0.0005 pF to 0.1758 pF). An analytical RC delay model is compared against SPICE results at the midpoint condition to assess first-order accuracy.

2. Characterization Results

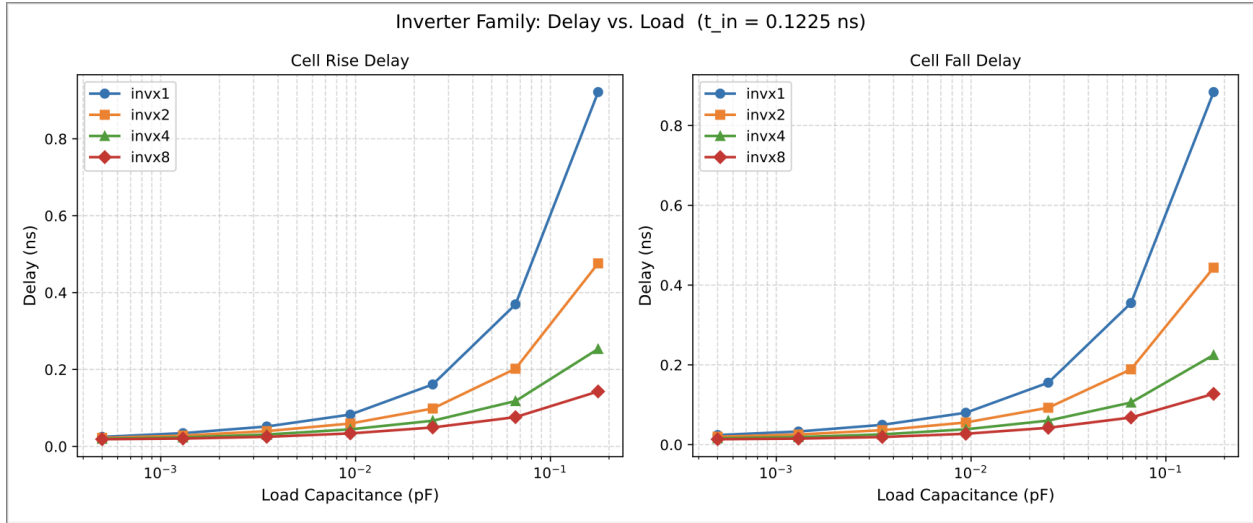
2.1 Inverter Family: Delay vs. Load Capacitance

Figure 1 shows the cell_rise and cell_fall delay versus output load capacitance for the four inverter drive strengths (invx1 through invx8), evaluated at a fixed input transition time of 0.1225 ns. The plot is on a logarithmic x-axis to span the full load range from 0.0005 pF to 0.1758 pF.

Key observations from the inverter family plot:

- Delay scales linearly with load capacitance across all drive strengths, as expected from the RC charging model.
- Each drive strength doubling (x1→x2→x4→x8) approximately halves the delay at any given load, confirming correct transistor width scaling.
- At very small loads (0.0005–0.001 pF), the delay is dominated by the intrinsic gate capacitance and the input transition time rather than the external load.

- Rise and fall delays are approximately symmetric for the inverter family due to the $2.38\times$ PMOS/NMOS sizing ratio.



2.2 NaN Entries

A small number of NaN entries appear in the tables at extreme conditions (invx8 and nand2x2/nand2x4 at $t_{in} = 1.5000$ ns, smallest loads). These occur when the PULSE period is shorter than the RC settling time at these large-drive, light-load conditions, causing the output not to complete its fall transition before the next rising edge. These represent operating points outside the intended characterization regime and are indicated as N/A in the tables.

3. RC Model vs. SPICE Comparison

3.1 RC Model Parameters

A first-order RC delay model was used to estimate the propagation delay of the invx1 cell.

The on-resistance was estimated using the linear region approximation: $R_{on} = 1 / (K_n \times (W/L) \times (V_{DD} - V_{tn}))$. The PMOS on-resistance was taken as $2 \times R_{on_n}$ to account for the lower hole mobility. The RC propagation delay was computed as: $t_{pd} = 0.69 \times R_{on} \times C_{load}$, which corresponds to the 50% crossing time for a step input driving an RC load.

3.2 Comparison at Midpoint Condition

The comparison was performed at the midpoint of the characterization table: input transition $t_{in} = 0.1225$ ns, load capacitance $C_{load} = 0.0094$ pF.

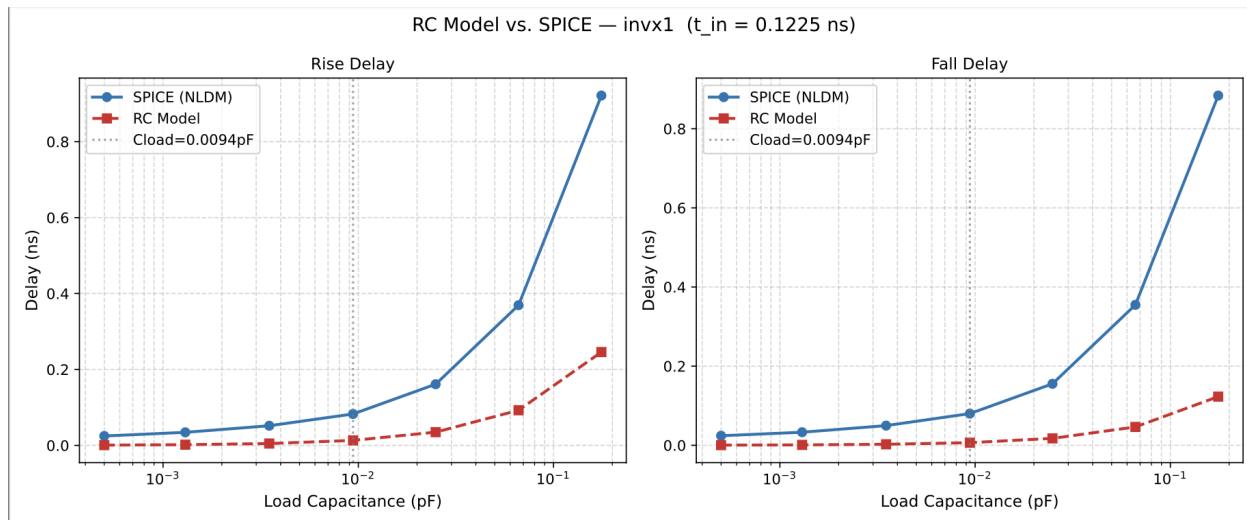
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RC Model vs. SPICE Comparison

Condition: $t_{in} = 0.1225$ ns, $C_{load} = 0.0094$ pF

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Metric	RC Model (ns)	SPICE (ns)	Error %
cell_rise (invx1)	0.0131	0.0827	-84.1%
cell_fall (invx1)	0.0066	0.0800	-91.8%



3.3 Discussion of RC Model Limitations

The RC model underestimates the SPICE delay by approximately 84–92% at the midpoint condition. This large discrepancy arises from several fundamental limitations of the first-order model:

Input transition effect (the primary limitation for this project):

The RC model assumes an ideal step input, a signal that transitions from 0 to V_{DD} instantaneously. In reality, the input transition takes 0.1225 ns at the midpoint. During this ramp, the MOSFET does not switch instantaneously: it passes through the saturation region (where the current is approximately constant at $I_{ds} = (K_n/2)(W/L)(V_{gs} - V_{tn})^2$) before entering the linear region (where R_{on} applies). The RC model uses only the linear-region resistance and completely

ignores the saturation-region current, which is the dominant charging/discharging mechanism during the early part of the transition.

The NLDM table captures this effect explicitly: looking at the `invx1 cell_rise` column at $C_{load} = 0.0094$ pF, the delay grows from 0.0562 ns at $t_{in} = 0.0100$ ns to 0.2488 ns at $t_{in} = 1.5000$ ns, a $4.4\times$ increase. The RC model, which ignores input transition entirely, produces the same 0.0131 ns estimate regardless of t_{in} . This is why NLDM tables are essential: they capture the two-dimensional dependence on both load and input slew that no simple closed-form model can represent accurately.

Velocity saturation and short-channel effects:

At $L = 0.15$ μm (short channel for a 180 nm node), SKY130 devices exhibit significant velocity saturation. The actual drive current is considerably lower than the long-channel square-law prediction used to compute R_{on} . This means the effective resistance seen by the load is higher than the RC model assumes, contributing to the underestimation.

Output capacitance and Miller effect:

The RC model accounts only for the external load capacitance C_{load} . It ignores the drain diffusion capacitance (C_{db}) of the output transistors, the gate-drain overlap capacitance (C_{gd} , amplified by the Miller effect during switching), and the input capacitance of the next stage. These internal capacitances can be comparable to C_{load} at the smallest load values (0.0005–0.001 pF), significantly increasing the effective load seen by the driver.

Body effect in stacked devices:

For NAND2 and NOR2 cells (and the internal nodes of MAJ3), the intermediate node of the stacked transistor pair floats during switching, raising the threshold voltage of the bottom device via the body effect. This further reduces drive current and increases delay beyond what the simple R_{on} model predicts.

Summary:

The RC model is useful as a first-order design tool for choosing transistor sizing and estimating drive strength trends, but it is inadequate for sign-off timing analysis. NLDM tables generated by full SPICE simulation accurately capture the nonlinear dependence on both input transition time and output load, making them the industry-standard format for static timing analysis.

Appendix: NLDM Tables for invx1

All values in nanoseconds. Index 1 (rows) = input transition time (ns). Index 2 (columns) = load capacitance (pF).

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Cell: invx1

cell_rise [invx1]
tin\cload(pF) | 0.0005 | 0.0013 | 0.0035 | 0.0094 | 0.0249 | 0.0662 | 0.1758
-----
0.0100 ns      | 0.0100 | 0.0146 | 0.0262 | 0.0562 | 0.1344 | 0.3425 | 0.8945
0.0231 ns      | 0.0130 | 0.0174 | 0.0291 | 0.0592 | 0.1375 | 0.3456 | 0.8977
0.0531 ns      | 0.0183 | 0.0243 | 0.0360 | 0.0662 | 0.1446 | 0.3528 | 0.9049
0.1225 ns      | 0.0247 | 0.0344 | 0.0517 | 0.0827 | 0.1611 | 0.3694 | 0.9215
0.2823 ns      | 0.0324 | 0.0471 | 0.0747 | 0.1199 | 0.1997 | 0.4078 | 0.9599
0.6507 ns      | 0.0418 | 0.0628 | 0.1043 | 0.1759 | 0.2880 | 0.4972 | 1.0488
1.5000 ns      | 0.0533 | 0.0822 | 0.1411 | 0.2488 | 0.4265 | 0.7051 | 1.2552

cell_fall [invx1]
tin\cload(pF) | 0.0005 | 0.0013 | 0.0035 | 0.0094 | 0.0249 | 0.0662 | 0.1758
-----
0.0100 ns      | 0.0109 | 0.0153 | 0.0265 | 0.0552 | 0.1301 | 0.3294 | 0.8582
0.0231 ns      | 0.0133 | 0.0177 | 0.0291 | 0.0580 | 0.1330 | 0.3324 | 0.8612
0.0531 ns      | 0.0181 | 0.0238 | 0.0354 | 0.0646 | 0.1397 | 0.3392 | 0.8681
0.1225 ns      | 0.0240 | 0.0329 | 0.0496 | 0.0800 | 0.1553 | 0.3549 | 0.8839
0.2823 ns      | 0.0308 | 0.0443 | 0.0700 | 0.1139 | 0.1917 | 0.3913 | 0.9203
0.6507 ns      | 0.0377 | 0.0575 | 0.0959 | 0.1632 | 0.2729 | 0.4758 | 1.0045
1.5000 ns      | 0.0421 | 0.0707 | 0.1268 | 0.2270 | 0.3947 | 0.6693 | 1.1998

rise_transition [invx1]
tin\cload(pF) | 0.0005 | 0.0013 | 0.0035 | 0.0094 | 0.0249 | 0.0662 | 0.1758
-----
0.0100 ns      | 0.0071 | 0.0131 | 0.0294 | 0.0732 | 0.1881 | 0.4945 | 1.3077
0.0231 ns      | 0.0077 | 0.0131 | 0.0294 | 0.0732 | 0.1881 | 0.4945 | 1.3077
0.0531 ns      | 0.0127 | 0.0164 | 0.0299 | 0.0732 | 0.1881 | 0.4945 | 1.3077
0.1225 ns      | 0.0205 | 0.0268 | 0.0385 | 0.0747 | 0.1881 | 0.4945 | 1.3077
0.2823 ns      | 0.0338 | 0.0442 | 0.0625 | 0.0936 | 0.1908 | 0.4945 | 1.3077
0.6507 ns      | 0.0570 | 0.0726 | 0.1025 | 0.1501 | 0.2310 | 0.4978 | 1.3077
1.5000 ns      | 0.1035 | 0.1239 | 0.1679 | 0.2460 | 0.3636 | 0.5792 | 1.3100

fall_transition [invx1]
tin\cload(pF) | 0.0005 | 0.0013 | 0.0035 | 0.0094 | 0.0249 | 0.0662 | 0.1758
-----
0.0100 ns      | 0.0058 | 0.0108 | 0.0247 | 0.0618 | 0.1595 | 0.4198 | 1.1105
0.0231 ns      | 0.0060 | 0.0108 | 0.0247 | 0.0618 | 0.1595 | 0.4198 | 1.1105
0.0531 ns      | 0.0093 | 0.0131 | 0.0250 | 0.0618 | 0.1595 | 0.4198 | 1.1105
0.1225 ns      | 0.0155 | 0.0214 | 0.0324 | 0.0632 | 0.1595 | 0.4198 | 1.1105
0.2823 ns      | 0.0265 | 0.0355 | 0.0521 | 0.0808 | 0.1623 | 0.4198 | 1.1105
0.6507 ns      | 0.0476 | 0.0604 | 0.0852 | 0.1283 | 0.2009 | 0.4238 | 1.1105
1.5000 ns      | 0.0914 | 0.1087 | 0.1438 | 0.2075 | 0.3145 | 0.5037 | 1.1143
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Complete NLDM tables for all 13 cells are included in the file results/nldm_report.md in the project GitHub repository.